Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/05/2025
PI Name:	Oza, Anand Ph.D.		
Project Title:	Phase Transitions in Colloid-Polymer Mixtures in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-E: Use of the NASA Physical Sciences Informatics System – Appendix E
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Assignment:	NOTE: End date changed to 11/03/2023 per NSSC information. (Ed. 11/21/22) NOTE: End date changed to 11/04/2022 per NSSC information. (Ed. 10/20/21)		
Key Personnel Changes/Previous PI:	no changes		
COI Name (Institution):	Khusid, Boris Ph.D. (New Jersey Institute of Technology) Kondic, Lou Ph.D. (New Jersey Institute of Technology) Meyer, William Ph.D. (Universities Space Research Association/NASA Glenn Research Center)		
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Task Description:	Adding non-adsorbing polymers to a suspension of colloidal spheres induces attractive forces between spheres originating from the presence of depletion zones around the spheres. We propose to work on analyzing the data on colloid/non-adsorbing polymer mixtures acquired in microgravity in NASA-supported projects Binary Colloidal Alloy Test (BCAT) 3-6. In our recently completed project (Structure evolution during phase separation in colloids under microgravity, NNX16AQ79G, PI: Kondic, 8/16/2016 - 8/15/2018) we analyzed the data on suspensions of hard spheres collected in Physics of Hard Spheres (PHaSE) experiments. As part of that project, we have formulated a new model that includes up-to-date information about the nature of phase transitions in suspensions of hard spheres. This model, coupled with the data acquired in the NASA-supported projects, already allowed us to explain some important observations of the crystallization kinetics in suspensions of hard spheres. These results were presented at the 2017 and 2018 American Society for Gravitational and Space Research (ASGSR) meetings. As a part of the proposed work, we plan to develop a new model for phase transitions in colloid-polymer mixtures. The model that we propose to develop will be based on the one that we developed for suspensions of hard-spheres, but will involve significantly more complicated physics. The predictions of the model will be compared to the experimental data residing in the NASA Physical Sciences Informatics (PSI) system, as well as to relevant Earth-based experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Colloids have been studied extensively because of the many similarities they share with behaviors of atoms in a microscopic setting. The fact that the colloidal particles are macroscopic makes colloids much easier to observe than atomic systems, and yet their behavior is such that they may be viewed as "big atoms." The addition of polymer into a colloidal mixture leads to the possibility of three-phase coexistence. This project thus has the potential to give us new fundamental insights into the phase transition kinetics of systems with complex phase behavior. Experiments with colloids have applications in fields such as materials science, engineering, and pharmaceuticals. Understanding phase transitions on an atomic scale is extremely useful in making and working with materials that are used to manufacture electronics and other products that we use in our everyday lives.
Task Progress:	Colloidal particles can be considered as blocks for building materials, like atoms for forming molecules and crystals. The recent surge of activity in assembly of colloidal crystals is motivated not only by scientific curiosity but also by myriad technologies in materials engineering, electronics, life science, and chemical and pharmaceutical industries. Colloidal systems undergo phase transformations analogous to transformations in atomic and molecular systems. Therefore, they are considered as excellent model systems for the study of phase transitions since the particle motions can be observed by optical video microscopy and tracked by image processing. However, the rich variety of colloidal structures observed in numerous terrestrial experiments could also have been influenced by gravity effects, such as particle sedimentation, convection, and jamming, which compete with colloidal forces during the slowly evolving structure formation. In this research program, we first process NASA Physical Sciences Informatics (PSI) video microscopy images to reveal the growth kinetics of colloid-polymer mixtures. We next conduct computer simulations of phase transitions in colloid-polymer mixtures under the conditions of microgravity experiments. Comparison of simulation results with data acquired from images recorded in microgravity experiments for over 5 orders of magnitude in time and more than 3 orders of magnitude in domain size provide for the first time the unique opportunity to rigorously test theoretical predictions for the kinetics of phase transitions. Analysis of these images has been used to develop new phase-field models that describe the phase separation process. The research results will provide recommendations for further development of theories, simulations, and ground-based experiments on phase transformations in advanced materials. We have processed NASA PSI video microscopy images to reveal the growth kinetics of colloid-polymer mixtures. We used the free software ImageJ to remove the extraneous background, adjust the brightness/contrast, and smooth the image, yielding a time-sequence of enhanced images that can be characterized quantitatively. We used the autocorrelation function to describe the time evolution of the length scale separating colloid-rich domains in the BCAT 3–5 experiments, and observed roughly linear growth of domains with time. We also developed a theoretical model for simulating phase separation in colloid-polymer systems. We use the Cahn-Hilliard-Stokes equation for the dynamics of the colloid volume fraction and the associated fluid velocity. Both are assumed to be functions of space in two dimensions and time. The dynamic of the colloid volume fraction is driven by the chemical potential, which is derived from a free energy functional. We assume that the free energy can be split into two parts: one corresponding to the colloid, and the other to the polymer, the latter modeling the weakly attractive interactions between colloids. The free energy also contains a gradient term that accounts for the diffuse nature of the interface between phases, as is typical in phase-field models. The velocity field is assumed to be determined by the incompressible Stokes equation forced by gradients in the colloid volume fraction through the so-called Korteweg stress. Our model differs from others in that the mobility is concentration-dependent; that is, the mobility decreases as the colloid volume fraction increases, and goes to zero as the crystal structure is approached. We expect this feature to significantly influence the simulated phase separation dynamics. Our model also differs from those previously proposed in that it accounts for viscosity variations in the Stokes equation; specifically, the viscosity is assumed to rise sharply with the colloid concentration. The role of hydrodynamic interactions in colloid-polymer phase separation has been the subject of controversy in the literature, so our simulations are able to shed light on the associated open questions. We have written a code in MATLAB for simulating the governing equations. The code uses a pseudospectral method in space and a semi-implicit time-stepping scheme based on splitting the mobility function. We validated our code against the results of a linear stability analysis, and performed convergence studies both in the time step and number of spatial grid points. Moreover, we have developed efficient algorithms for solving the variable-viscosity Stokes equations, which involves solving a large system of linear equations that determine the fluid velocity at every time step. Numerical simulations of the governing equations show that the system does indeed exhibit phase separation, with colloid-poor areas formed amongst a background of colloid-rich ones. The colloid-poor areas become more circular as time goes on, and the system coarsens, in that these areas merge with nearby ones. We find that such merger events are correlated with dramatic increases in the fluid speed, and are associated with short-lived quadrupolar vortex structures that surround the merging domains. We find that including hydrodynamics causes the domains to become more circular, while omitting hydrodynamics encourages the formation of more elongated structures. Moreover, including hydrodynamic interactions improves the agreement with experiment of the predicted rate of domain growth.

	The results have been disseminated through presentations at international scientific conferences and a scientific manuscript. Specifically, the PhD student on the project, Lauren Barnes, presented posters at the American Society for Gravitational and Space Science Research (ASGSR) Meetings in Baltimore, MD in November 2021 and Houston, TX in November 2022, and gave remote oral presentations at the ASGSR 2020 meeting and American Physical Society (APS) Division of Fluid Dynamics meeting in November 2021. The Principal Investigator (PI) was invited to give a talk on this work in the invited session, “Soft Matter in Low Gravity” at the American Physical Society (APS) March Meeting in Chicago, IL in March 2022. The PI also gave an oral presentation at the ASGSR 2023 Meeting in Washington, D.C., and gave an invited talk on this work at the International Congress on Industrial and Applied Mathematics (ICIAM) in August 2023. The PI, co-PIs, and PhD student have co-written a manuscript entitled, “Phase-field modeling of colloid-polymer mixtures in microgravity,” which is currently under review at the open access journal npj Microgravity. The manuscript is freely accessible on arXiv to encourage timely feedback from the scientific community, and to increase the visibility of the work.
Bibliography Type:	Description: (Last Updated: 03/24/2025)
Abstracts for Journals and Proceedings	Oza AU, Barnes L, Kondic L, Khusid B, Meyer WV. "Phase-field modeling of colloid-polymer mixtures in microgravity." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. https://asgsr.org/2023-meeting/ , Nov-2023
Articles in Other Journals or Periodicals	Barnes L, Khusid B, Kondic L, Meyer WV, Oza AU. "Phase-field modeling of colloid-polymer mixtures in microgravity." arXiv preprint server. Posted December 17, 2024. https://arxiv.org/abs/2412.12777 , Dec-2024
Articles in Peer-reviewed Journals	Swinkels PJM, Gong Z, Sacanna S, Meyer WV, Schall P. "Colloidal molecules in microgravity assembled by critical Casimir forces." Gravit Space Res. 2025 Mar 13;13(1):21-9. https://doi.org/10.2478/gsr-2025-0001 , Mar-2025